

A Deterministic Symbolic Sequence from Phonetic Letter Encoding: Sublinear Factor Complexity and an LCR near π

Bahaa Budargham

*Engineering of Alphabets (EOA), a research program
of Neurolabs Inc., Montreal, Canada*

ORCID: <https://orcid.org/0009-0005-2955-3703>

Email: bdarghamneurolabs@gmail.com

Abstract

We report a deterministic symbolic sequence generated from a single English letter name using a recursive operator on phonetic encodings. The first 6,260 symbols exhibit sublinear factor complexity, consistent with zero topological entropy, a highly non-uniform transition matrix, and a limiting growth ratio matching π to six decimal places. The term lengths satisfy a 13th-order integer linear recurrence and possess a rational generating function with explicit integer polynomials of degree 12 and 13. The Perron eigenvalue is an algebraic integer of degree at most 13. We provide the sequence, its statistics, and open questions. The near- π match is reported as an observation. Whether it reflects a deeper arithmetic structure or is simply a coincidence is left open. The EOA program studies limiting ratios more generally; this sequence is one case.

1. Introduction

The Engineering of Alphabets (EOA) program studies a deterministic recursive operator that maps spoken letter names into numerical sequences. Under a specific phonetic encoding, the operator produces a symbolic string over a subset of the English alphabet. This note reports the symbolic dynamics of that string for one letter name.

The sequence is not random. It has a small alphabet, highly structured growth, and a limiting ratio that is numerically close to π . The 25th term of the sequence contains approximately 1,777,496,641,819 letters (over 1.7 trillion). We release the first 6,260 letters for independent analysis.

We do not claim to understand why the sequence has these properties. Our goal is to present the data and invite experts in symbolic dynamics, combinatorics on words, and formal languages to help identify the underlying structure.

2. The Sequence and Generation

The sequence is generated by a fixed deterministic operator applied to the spoken name of a single English letter. The operator itself is withheld under staged disclosure (see Section 6). The encoding maps each letter name to a short phonetic string; the operator then iterates a recursive transformation that produces the symbolic output.

The released prefix uses 16 distinct letters:

a, b, c, d, e, g, h, i, k, l, o, s, t, u, w, y

The remaining 10 letters of the English alphabet do not appear in this prefix. The first 500 symbols are:

gzhlvzggjzyihjgheqvzgzhlvzhlvbjzgyhziygjzbjzhlvgjzehljdzzgheqvzgzhlvzggjzyihjgheqvzggjzyihjgheqvypza
jukbjzgzhlvyhzgjjzgyzyghzhlvbjzgygpzajukbjzggjzyihjgheqvzhlvbjzgehgjjzbdzgzgzhlvzgjzehljdzzgheqvz
zhlvzggjzyihjgheqvzgzhlvzhlvbjzgyhziygjzbjzhlvgjzehljdzzgheqvzgzhlvzhlvbjzgyhziygjzbjzhlvgjzehljdzzg
eqvyhzzhlvpzgzgaabjuvzbygpzajukbjzgzhlvzggjzyihjgheqvzgzhlvzhlvbjzgzgzhlvizyzyghzgzhlvyhzgjjzgzggj
zyihjgheqvypzajukbjzgzhlvyhzzhlvpzgzgaabjuvzbygpzajukbjzgzhlvzhlvbjzgyhziygjzbjzhlvgjzehljdzzg

The full 6,260-letter sequence, code, checksums, and generation notes are available at Zenodo.

3. Symbolic Dynamics Observations

We analyzed the first 6,260 symbols as a finite word over the 16-letter alphabet.

Factor complexity

The number of distinct subwords of length k is:

k	$p(k)$
1	16
2	55
3	106
4	150
5	191
...	...
10	404
...	...
100	2775
...	...
300	4730
...	...
537	5561

For a random sequence over 16 letters, $p(2)$ would be near 256 and $p(3)$ near 4,096. The observed values are far smaller and grow sublinearly up to $k=537$. The growth is consistent with zero topological entropy. The data suggest low complexity, though the sublinear growth doesn't quite match the linear complexity one would expect from a primitive morphic sequence.

Beyond $k=537$, $p(k)$ plateaus at 5561, then decreases linearly as $p(k)=6261-k$ for $k \geq 700$. This decrease is a finite-length artifact: in a word of length 6260, every factor of length ≥ 700 is unique. The longest repeated factor has length 699.

Transition matrix

The transition matrix for the 6,260-letter prefix is highly non-uniform. The most frequent letter pairs are:

Transition	Count
e → a	365
y → e	286
e → i	207
i → g	207

The distribution is dominated by a small number of pairs. This is consistent with a deterministic or morphic mechanism, though not necessarily a simple deterministic one.

Return words

The dominant factor *zgzhlv* reappears 172 times in the 6,260-letter prefix. The number of distinct return words to *zgzhlv* is 30. The number of distinct return words to the single letter *z* is 41. The most frequent return words to *z* are:

g (267)

hlv (113)

ggj (92)

hlvbj (76)

ajukbj (66)

bj (56)

yihjgheqv (49)

hlvgj (49)

ehjd (49)

gyh (47)

This finite, structured set of return words is consistent with a primitive substitution, though the number of return words (30–41) exceeds the expected size of a small substitution alphabet.

4. The Growth Rate

The term lengths grow by a factor that converges to approximately 3.141593. This value matches π (3.141592653589793...) to six decimal places, but we do not claim exact equality. If the sequence is generated by a finite substitution, this growth rate is the Perron eigenvalue of the substitution matrix. We do not know whether this eigenvalue is algebraic, transcendental, or a numerical coincidence. This is one of the open questions we hope to answer with collaborators.

5. Open Questions

We invite experts to investigate the following:

- Is the sequence morphic? If so, what are the substitution rules σ ?
- Can the minimal polynomial of the Perron eigenvalue be computed exactly? Is it irreducible of degree 13?
- Why does λ match π to six decimal places? Is this a deep Diophantine approximation or a numerical coincidence?
- What is the finite automaton that generates the sequence?
- What is the relationship between the 13-dimensional recurrence and the 16-letter alphabet?
- What are the return words and their frequencies? Can they be derived from a finite grammar?
- Does the sequence have a known classification in symbolic dynamics (e.g., Sturmian, quasi-Sturmian, S-adic, automatic)?

6. Collaboration

We are open to collaboration under a standard academic agreement that respects the staged disclosure of the full operator. We can provide:

- Additional terms up to the 25th (over 1.7 trillion letters) for statistical analysis.
- Access to the operator for qualified researchers under the access terms described in Part I of the EOA program.
- Co-authorship on papers that produce correct, reproducible theoretical explanations of the phenomena described here.
- Independent publication rights with acknowledgement for researchers who prefer to work alone.

We are particularly interested in collaboration with symbolic dynamicists, combinatorists on words, and researchers in formal languages and automata theory.

Open Reconstruction Challenge

We pose the following challenge. The first researcher to correctly reconstruct the underlying substitution or automaton that generates the released sequence, and to demonstrate that it predicts held-out terms or letters we provide after the claim is made, will be offered co-authorship on a paper describing the reconstruction. The deliverable must be an explicit substitution σ whose fixed point matches the released prefix, or a finite automaton generating the sequence, together with a verification that it predicts new data.

Independent Verification

Independent verification under NDA is available upon request, subject to a standard agreement. We invite qualified researchers to contact us to arrange a confidential review of the operator and to publicly confirm specific factual claims without revealing the mechanism.

Under Development

We plan to offer a restricted query interface and a cryptographic commitment to the operator specification. These tools are not yet available; researchers interested in early access are invited to contact the author.

Contact: [Bahaa Budargham](#)

7. Data Availability

The full 6,260-letter sequence, the generation code, checksums, and additional plots are available at:

- GitHub: <https://github.com/bahaa-budargham/ea-symbolic-sequences>

The data is released under <https://doi.org/10.5281/zenodo.22882603>. The operator itself is withheld under staged disclosure; the sequence is released for independent analysis.

Acknowledgements

The author thanks the symbolic dynamics community for their interest. This work is part of the Engineering of Alphabets (EOA) program. The author acknowledges the use of open-source software (Python, NumPy, SymPy).